

# Analyst's Case File

CTS 285 · Systems Analysis and Design · Module 1

|                  |                                          |
|------------------|------------------------------------------|
| System analyzed  | DataMan (Texas Instruments, 1977)        |
| Feature assigned | Number Guesser                           |
| Source document  | The Story of DataMan — manual transcript |
| Your name        | Elijah Dixon                             |
| Date             | 9/23/2026                                |

## How to fill this out

Every finding needs evidence. Evidence is an exact quoted phrase from the manual plus the section heading it came from. Cite sections by name — “Number Guesser (Story)” or “Electro Flash · Number Guesser · Wipe Out (Operating Notes).” Do not cite page numbers; the manual carries two numbering systems that do not agree.

Read both registers before you write anything. Part I describes the feature to a child. Part II describes it to an adult. Each contains information the other omits.

If you cannot find evidence for something, do not guess. Put it in Section 8. An unknown you can name is a finding. An assumption you record as fact is an error.

*Rows are a format, not a quota. Add rows as your evidence requires.*

## 1. Purpose

Why does this feature exist? What problem does it solve for the user?

| Finding                                                                                             | Evidence (exact quote)                                                                                                                                                                                                                                                                                                           | Section                                                     |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| To get children to become more familiar with numbers by engaging with them in a fun way.            | “One thing that annoys AntiMath and makes him green with envy is people having fun with numbers and mathematics!” <b>(Story)</b><br><br>“As you play Number Guesser with your family or friends (or just by yourself), you'll be sharpening your math skills. AntiMath will keep his distance whenever you play!” <b>(Story)</b> | Number Guesser (Story)                                      |
| To teach children about the concepts of “betweenness”, and number logic (estimation and averaging). | “Number Guesser helps to teach the important basic concept of number betweenness ...” <b>(Op. Notes)</b><br><br>“The strategy involved in getting to the answer in the fewest tries will help build skills in estimation and averaging.” <b>(Op. Notes)</b>                                                                      | Electro Flash · Number Guesser · Wipe Out (Operating Notes) |

| Finding | Evidence (exact quote) | Section |
|---------|------------------------|---------|
|         |                        |         |

## 2. Users and Roles

Who interacts with this feature? Where more than one person is involved, name what each one does — the person who sets something up and the person who responds to it are different roles even when they are the same human being.

| Role              | What this role does                                                                                                  | Evidence (exact quote)                                                                                                                                                                                                                                           | Section                                                           |
|-------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Family            | Play the game with the individual operating the device.                                                              | <p>“As you play Number Guesser with your family or friends (or just by yourself), you'll be sharpening your math skills...” <b>(Story)</b></p> <p>“Number Guesser is an educational game of number strategy the whole family will enjoy.” <b>(Op. Notes)</b></p> | Both Sections                                                     |
| Friends           | Peers of the individual using the device.                                                                            | <p>“As you play Number Guesser with your family or friends...” <b>(Story)</b></p> <p>“Take turns with your friends trying to guess the secret number in the fewest tries.” <b>(Story)</b></p>                                                                    | Number Guesser (Story)                                            |
| “Yourself”, “You” | Refers to the individual operating the device. Might be the child, a friend, or a family member.                     | <p>“As you play Number Guesser with <u>your</u> family or friends (or just by <u>yourself</u>)...” <b>(Story)</b></p> <p>“As <u>you</u> enter each guess...” <b>(Op. Notes)</b></p>                                                                              | Both Sections                                                     |
| Children          | <p>Refers to the target users of the device.</p> <p>(eg, a parent may be using the device with a child/children)</p> | <p>“Children can take turns seeing who can find a secret number in the least guesses!” <b>(Op. Notes)</b></p>                                                                                                                                                    | Electro Flash ·<br>Number Guesser ·<br>Wipe Out (Operating Notes) |

## 3. Inputs

What information or action enters the system? Include keystrokes, choices, and anything the system refuses to accept.

| Input                 | Supplied by | Evidence (exact quote)                                                                                           | Section |
|-----------------------|-------------|------------------------------------------------------------------------------------------------------------------|---------|
| Turning the system ON | ON key      | <p>“Just press ON...” <b>(Story)</b></p> <p>“...first turn DataMan on with the ON key...” <b>(Op. Notes)</b></p> | Both    |

| Input                | Supplied by              | Evidence (exact quote)                                                                                                            | Section       |
|----------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|
| Start number guesser | [??] NUMBER GUESSER key. | “...and the Number Guesser key: [??]...” <b>(Story)</b><br><br>“...and then press the [NUMBER GUESSER] key...” <b>(Op. Notes)</b> | Both Sections |
| The users guess.     | Number keys, “=” key.    | “You enter your guess...” <b>(Story)</b><br><br>“As you enter your guess...” <b>(Op. Notes)</b>                                   | Both Sections |
|                      |                          |                                                                                                                                   |               |

## 4. Processing

What does the system do with the input? Include any rule, limit, or constraint you can support.

| Rule or behavior                                                           | Evidence (exact quote)                                                                                                                                                                                                                                                                                      | Section |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Dataman chooses a number between 9 and 100, after the [??] key is pressed. | “Just press ON and the Number Guesser key: [??] I'll pick a secret number for you to guess. It will be somewhere between 9 and 100.” <b>(Story)</b><br><br>“...and then press the [NUMBER GUESSER] key. Then, try to guess a secret number between 9 and 100 that DataMan has Selected.” <b>(Op. Notes)</b> | Both    |
| For each guess, the dataman will display a hint.                           | “...DataMan provides a hint by displaying two numbers that the secret number is between.” <b>(Op. Notes)</b><br><br>“When you do, I'll flash and show you two numbers in my face mask.” <b>(Story)</b>                                                                                                      | Both    |
|                                                                            |                                                                                                                                                                                                                                                                                                             |         |
|                                                                            |                                                                                                                                                                                                                                                                                                             |         |

## 5. Outputs

What information does the system return to the user?

| Output | When it appears                                                      | Evidence (exact quote)                                                                                       | Section |
|--------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------|
| Hints  | When starting the game initially, and is refreshed after each guess. | “...DataMan provides a hint by displaying two numbers that the secret number is between.” <b>(Op. Notes)</b> | Both    |

| Output     | When it appears                   | Evidence (exact quote)                                                                                                                                                                           | Section |
|------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|            |                                   | "When you do, I'll flash and show you two numbers in my face mask." <b>(Story)</b>                                                                                                               |         |
| Light Show | After the correct guess is given. | "...and then a great light show!" <b>(Story)</b><br><br>"When the secret number is found, DataMan rewards you with a spectacular 'light show'." <b>(Op. Notes)</b>                               | Both    |
| Guesses    | After the correct guess is given. | "When you finally guess the secret number, I'll show you the number of tries it took..." <b>(Story)</b><br><br>"...and displays the total number of guesses that were taken." <b>(Op. Notes)</b> | Both    |

## 6. Feedback

How does the system respond to the user's actions? Feedback tells the user how they are doing. An output returns information; feedback comments on it. If you think one behavior is both, record it once and say why.

| Feedback behavior                             | What it tells the user | Evidence (exact quote)                                                                                                                                                                           | Section |
|-----------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Light display.                                | Correct Guess!!        | "...and then a great light show!" <b>(Story)</b><br><br>"When the secret number is found, DataMan rewards you with a spectacular 'light show'." <b>(Op. Notes)</b>                               | Both    |
| Displays the number of guesses to the screen. | Number of attempts.    | "When you finally guess the secret number, I'll show you the number of tries it took..." <b>(Story)</b><br><br>"...and displays the total number of guesses that were taken." <b>(Op. Notes)</b> | Both    |
|                                               |                        |                                                                                                                                                                                                  |         |

## 7. Observations and Assumptions

List three statements you were tempted to write as fact but could not fully support. For each, say what evidence you would need.

| Statement                           | Observation or assumption? | What evidence would settle it?                                                   |
|-------------------------------------|----------------------------|----------------------------------------------------------------------------------|
| The Number Keys as an input option. | Assumption (partially?)    | Though there are instructions that mention the number keys in other parts of the |

| Statement                                                                                                   | Observation or assumption? | What evidence would settle it?                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                             |                            | <p>instructions, the Number guesser sections do not mention them directly.</p> <p>They only say to guess the numbers, though you could argue it is implied the user would know how to input numbers as you would have read the earlier sections of the manual.</p> <p>If the instructions directly said “guess the number by using the number keys and input you answer using the = button”, then it wouldn’t be an assumption.</p> |
| Only non-negative numbers between 9-100 allowed for input.                                                  | Assumption                 | The instructions for Number guesser never give any restrictions. Other sections do, but by reading this alone it would be an assumption to state this as a constraint.                                                                                                                                                                                                                                                              |
| Number Guesser’s hints will get within an increasingly narrow range of the random number chosen by DataMan. | Assumption                 | Normally, when playing a number guessing game with real people you would make the range smaller and smaller. Rather than just providing 2 random numbers. Without having the actual device to confirm this, it would be an assumption to believe this.                                                                                                                                                                              |

## 8. Unknowns and Open Questions

What could not be determined from the manual? For each, state what you looked for and where you looked, so a reader knows the gap is real and not merely unsearched.

| Open question                                                                                                    | Where I looked                                                                                | Why the manual does not answer it                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Do the numbers given as hints have any sort of pattern? Does the range narrow in relation to the number guessed? | In the Number Guesser (Story) and Electro Flash · Number Guesser · Wipe Out (Operating Notes) | It may be that the novelty of the game comes from not knowing how this is calculated. Or there might be more information that the user would know from using the device. |
|                                                                                                                  |                                                                                               |                                                                                                                                                                          |
|                                                                                                                  |                                                                                               |                                                                                                                                                                          |

## 9. Cross-Register Note

This manual documents the same feature twice. Identify one thing your assigned feature’s two sections handle differently — something one section states and the other omits, or something the two describe in ways that do not match.

### What differs:

The operating notes mentions specific skills like “estimation” and “averaging”, while the story generalizes them as “math skills”.

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**Part I says (quote):**

“...you'll be sharpening your math skills.”

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**Part II says (quote), or is silent:**

“The strategy involved in getting to the answer in the fewest tries will help build skills in estimation and averaging.”

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**Why this matters to an analyst:**

Knowing why the operating notes mention this detail and not the story section is important! The device will be interacted with/ sold to two audiences: children and their parents/guardians. Children want to have fun with the device, while parents want their children to learn new math skills.

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## Before you submit

- ☐ Every finding has a quoted phrase, not a paraphrase.
- ☐ Every quote names the section it came from.
- ☐ I cited section headings, not page numbers.
- ☐ I read both Part I and Part II before writing.
- ☐ Section 8 is not empty.
- ☐ Nothing in Sections 1–6 is a conclusion I could not support.